



Getting Started with Matplotlib Transcript

Welcome to the lesson number 21. In this we will be trying to demo the matplotlib features in a collab environment. As I said I will be simultaneously exhibiting the code in a live collab book so that you can go back and see how these features are working while practising using the standard template that I will be providing here.

Okay, so we have discussed the features and the benefits of matplotlib. So the task here is today to create a simple plot of sine wave which you know would oscillate between 0 to 1 and has a nice curvature which goes ups and downs. We will try to plot this.

So for that as I said any function that you want to do in a Python environment requires certain libraries. So in this case we need the matplotlib library. We will be importing the necessary libraries into the code environment.

So we use the import function and call the library name matplotlib.pyplot as plt. This is the call name and also you have the number as np which for being mathematical calibrations and this particular line dot matplotlib inline helps you to produce the results in the same environment. Otherwise it generally takes you to another window so that the graphs and the visualisations that are created are within the same notebook so that you can readily see the result.

I am just running this code. So yeah sometimes if there is an error in this you can always ask Gemini to help you understand the error and try to fix it also. Okay and as I said our task here is to create the sine wave.

So first I am trying to prepare the data. Okay so yeah so I'm just trying to pop the first two elements. As I mentioned the hash if you use hash before any line it would be annotated not as code and some instruction or some text element for the person who is either making the code or asking others to read or edit this code.

These are guidance lines. These are not code elements. So if you don't annotate the line it might show an error.

So always remember if you want to show some description or annotations please use the hash symbol to remove that as part of the code. So now I am using the features so defining X variable X which has a linear space then which is a sequence of linear numbers. So see it is also if you click on this you are getting some description of the function.

So it is also you need not go back and look at. So it says what is the function. So it gives you evenly spaced numbers calculated over a interval.



What is the interval 0 to 100 and it is sequence of 10 steps right. So you there are different ways to you can do this. So there are an exam I think they will give you an example also.

So you can always try to understand what is the function and use it for the features. Similarly sine X function so you can select the. So I am defining another variable Y as sine X sine X function right.

Then I am just running these two. Then in this now so I am just now trying to create a plot. So while you are creating a plot in math plot in the math plot environment you use the figure function and define the PLT dot subplot is the function and you are defining it to be AXE right.

So if you select this so you get the description of the function. So you can understand what is it trying to do okay. So it shows create a figure and a subset of subplots.

So we will see the functions in more more detail as we explore different examples. So we are just defining now for this sine X function. So this is the figure that I want to create okay.

So now what am I doing I am also adding the plot in the data AXE plot. So I have defined the variables I have defined the figure area. Now I am defining the I am asking the math plot the Python to plot this AXE dot plot plot function using X variable and Y variable and label it as sine right.

So see clearly the sine function is plotted starting from 0 right. Then up till 10 it has plotted and it has shown. So now what I want to do this is where our main element is we want to customise the plot.

It is our objective is not just to draw the plot but to customise why visualisation is all about bringing creative attributes into the environment and make it more visually appealing or easily discernible to our audience okay. We will always look at how better to present the data. So what am I doing we are using axis dot set underscore title which will give the title to the chart.

So simple sine wave and it will say this as the bottom horizontal as x-axis so set X label as x-axis Y label as y-axis okay. So this is what we have done and then see this is the intelligent feature in the collab environment it is trying to show the next features it is asking do you want a legend right or you want to finally show the plot. So actually the show plot function is not required here but it is just in case if you are running you want to actually are running the code you want to show the plot you have to take the plot function show plot function right.

So we are doing this plot dot show just for completeness. So how is this working first I am defining the X variable as a linear space then Y as a sine of Y is equal to sine of X Y variable as sine of X and I am calling the subplot function to define a figure then I am using the AXE dot plot function to plot the graph and I am customising them by adding the title labels to the axis and finally showing the diagram. So this is how you have got now the thing is you have got x-axis you have got y-axis and you have got the simple single.

Now I just want to a bit complicate this a little further so what I am trying to do I am trying to add one more one more line to the plot I want to simultaneously plot the cos X function also in this sine X and then the cos X function. So I am saying this is the cos X this is Y variable is sine X this is cos X and what am I doing subplots the same thing so I am using the another function here right let me take the description also so that we want to show this cos X function on the same subgraph so you are already plotted the sine wave we are going after this so we are just un-commenting it. So look at the function AXE dot plot for the same X it is using Y2 instead of Y and colour is red you can view colour is red line style is double hat it is not a connected line it is dashed line then you have the cosine label is cosine so you are labelling it as cosine okay cos so then finally we have the x-axis y-axis is same red is true right so we also now want to show the legend because you want to understand which is what right so now you have the two functions simple sine wave and cos wave so cos wave is different sine and cos are shown legend is also now present right and now you want to add the grids for better readability or remove grids for better readability if you have the grids so I am also including the grids so once we do this you have the grids for better readability so this is all can use the simple features in matplotlib for understand for building certain simple graphs now you can save the final see after developing the graph you want to save it so that you can import it to your PPT or your word document in which you are making the final presentation okay so you can use this function fig dot save okay save fig my plot dot PNG and we get saved in the local in your desktop or your lab right now we will now use these features for developing one more exercise example here you have to plot a linear line in the developer graph for a linear line so this is the task so I'm just popping the task formalities so let us understand what is the task so you have to plot the okay what is the mistake I have done I have pasted this into the code environment that's how it is showing it is highlighting all the functional elements and if I just run this right so it shows syntax error so be very careful so you can toggle it to yeah so you just make it on and delete the cell so you can just delete the cell and create a text cell and paste it so now I just do this so you have to plot a function y is equal to 2x plus 5 which is a linear length so we are trying to generate values of x from 5 minus 5 to 5 then define y is equal to 2x plus 5 then add a plot function and try to plot so how to go about this so you simply insert a code element first of all try to define the variables so same command like last time so we will be using y is equal x is equal to a linear function linear spaced variable minus 5 to 5 okay it will come from minus 5 to 5 right and defining y is equal to 2 times of x plus 5 I just run so the variables are

now set then I am trying to use the again so so it's subplots function for defining the plots so fig.

I am using the subplot functions which is create the initial plot space in which your figure will come then I am using the plot function to plot x comma y and giving colour green so we can also give colour blue you can directly or the red for that matter you can directly give the colours we will look about colours in the next lesson but just so see the moment I just typed the intelligent agent has given certain suggestions set title is equal to linear function y is equal to 2x plus 5 set x label by label and for this time I'm just trying to do the tab I just press the tab accepting the suggestions given by the intelligent agent and doing the plot functions when you want this right so now let us look at the plot function more closely so

so what we can do in the plot function we can also include the line style so I'm just going to try this here try this out line style see this is the interesting part so once you type half of the function name you get a call-out box explaining what are the different features that are there in that so that you can use understand the functionality and then use that okay so yeah so yeah line state so I'm going to new line status is equal to double point no one to do dot deadline let me see if it works it is not working symbol bash symbol bash is this then boss so just try out the way you learn is by doing mistakes and trying to correct the code as you go okay this brings us to the end of this lesson

And before I exit I just want to take you to certain examples that are there in math plotlib so let us talk about lines bars okay there are many many examples here the list is endless it is your patience and it is your so the best way I would say the best practises learn certain common minimum elements so if you recall in our second lesson we said a second or third lesson we said in the first module there are about 13 to 14 standard graphs that are used again and again in across all visualisations which would serve about 90 to 95 percent of your tasks at least for those graphs you should be thorough in using Python so that you can customise them so it is like this most of your data visualisation tasks will be solved by about 10 to 15 graphs or 10 to 15 visualisations and these visualisations you should be able to use develop this visualisation using Python environment and within that environment you should be able to customise them as you want so this is the I think in some sense the broader learning objective of this module where in those standard graphs you are able to customise as you want using Python environment so let us simply take the this infinite lines is one example then then you have the bar chart right with different categories so I'm just trying to do one thing I'll just take this code okay to the Colab environment right and try to see and try to explain the code to you okay so what is it doing a PLT dot subplots it will create the plot figure space right so I will sequentially see we have already imported the matplotlib as PLT there is no need to run that again or yes so I am doing just run this selection so the only selected part of the code will be run so once you do the PLT dot subplots and define this as X is equal to this is the call function

so then this is the figure so it shows the figure in which you are going to create the visualisation right then we are defining variables like an array in this case array is nothing but a collection of objects not getting into the technical it is you can get it or you can learn that from the introduction session or some Python basics so fruits is equal to apple blueberry and cherry and orange and counts number of foods is equal to 40 100 30 55 here there is a correspondence between each of the array element then you have bar with labels which are red blue underscore red and then orange so you want to then have bar colours tab red tab blue red and tab orange so now what are you doing so the point here is let us try to go one by one so you are having fruit you are having count a simple bar function could be let us now just run these two first let me run the selection okay I've done this okay then I am trying to run a simple bar graph okay so I am just doing bar graph without these two okay I'm just running a simple bar with fruits and count and selection okay so yeah so what has happened I have just defined two arrays fruits and counts so I am just taking apple blueberry cherry and orange right and then but this is not very pleasing we want to add some pretext elements and preattentive attributes like colour to this chart so I'm just using the labels I am trying to create two arrays I want to colour each fruit as different and then I also want to get the colours different for this and I want to add labels to each of these colours okay so then I am using these two and saying food supply I'm adding the title and also adding the legend and let me now run the search yeah so let me just run this for you so this is the first graph where you had the bland blue colour for all the variables then you are trying to do something different so this is the difference see so you are able to write the you are able to give the labels red blue right then you got the fruit colour then you have got the different colours for the bars right and you are able to food supply by kind and colour right then you are labelling it so this is a much more visual visually appealing graph than the other and then you are just using these features the key here is to create an array of bar labels and bar colours you have created an array and then you are able to use that in the graphic okay yeah now this is a basic intro what I would add all of you is to go through the documentation and the examples the port is given here then you have the Python book and the data so you can practise it with your in your free time and I sincerely suggest each one of you are to go deeply into these examples and develop your confidence thank you

-----End-----